

WEEK 4 ASSIGNMENT

Azure Data Factory End-to-End Data Pipeline Using Azure Blob Storage

Objective

The objective of this assignment is to understand Azure cloud services and implement an end-to-end data integration pipeline using Azure Storage Account and Azure Data Factory (ADF). The project demonstrates data ingestion, metadata validation, data movement, monitoring, and access management within the Azure ecosystem.

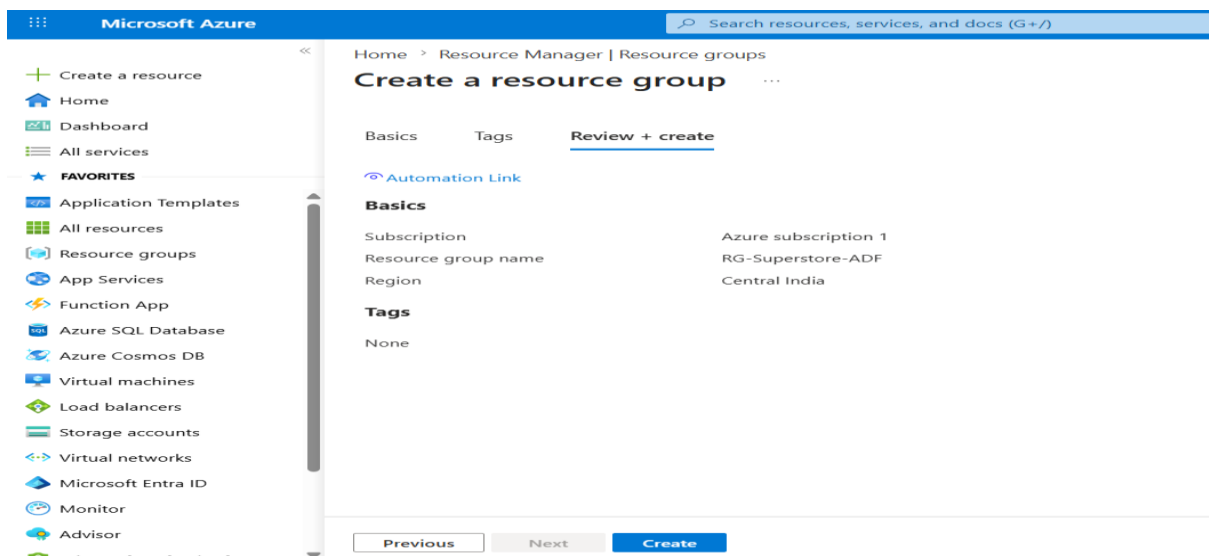
Environment Details

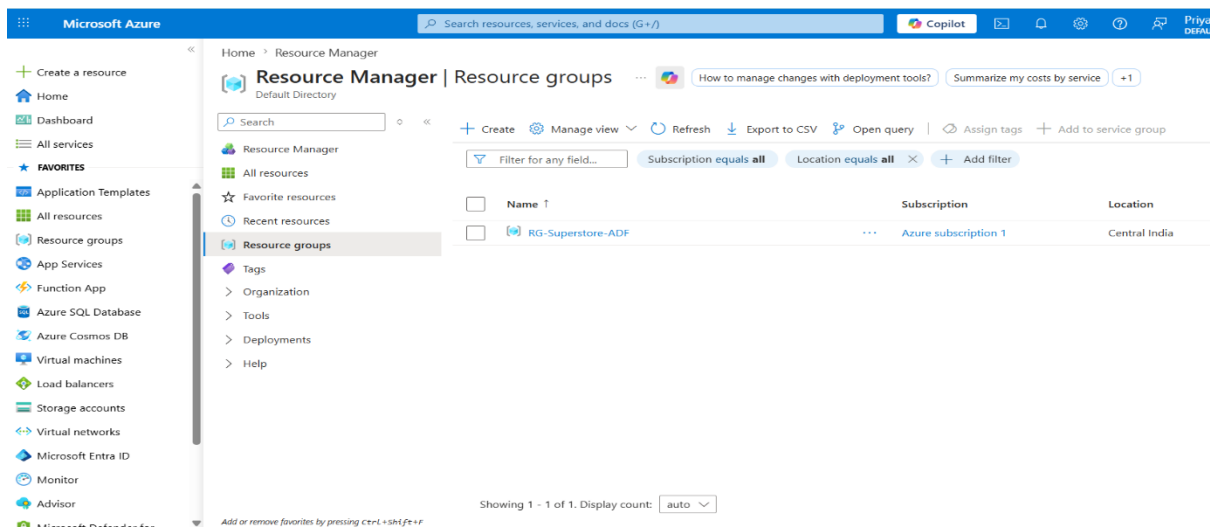
Service	Azure Resource
Cloud Platform	Microsoft Azure
Storage Service	Azure Blob Storage
Integration Service	Azure Data Factory
Dataset Used	Sample Superstore.csv
Source Container	superstore-data
File Format	CSV

Step 1: Resource Group Creation

A Resource Group was created in Azure Portal to organize and manage all resources associated with the project.

Output





Step 2: Storage Account and Blob Container Setup

An Azure Storage Account was created to store project data.

A Blob Container named **superstore-data** was created and the **Sample-Superstore.csv** file was uploaded successfully.

Output

The image shows two screenshots of the Microsoft Azure portal. The top screenshot displays the 'Create a storage account' wizard, and the bottom screenshot shows the 'Storage center | Blob Storage' resource page.

Top Screenshot: Create a storage account

The 'Create a storage account' wizard is shown with the 'Review + create' tab selected. The 'Basics' tab is active, showing the following configuration:

Property	Value
Subscription	Azure subscription 1
Resource group	RG-Superstore-ADF
Location	Central India
Storage account name	superstoreadf2026
Primary service	Azure Blob Storage or Azure Data Lake Storage
Performance	Standard
Replication	Locally redundant storage (LRS)

The 'Advanced' tab shows the following configuration:

Property	Value
Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled

The 'Review + create' tab is selected, showing the 'Previous', 'Next', and 'Create' buttons.

Bottom Screenshot: Storage center | Blob Storage

The 'Storage center | Blob Storage' resource page is shown. The 'Resources' tab is active, displaying a table of storage accounts.

Name	Type	Kind	Resource Group	Location	Subscription
superstoreadf2026	Storage account	StorageV2	RG-Superstore-A...	Central India	Azure subscriptio...

Step 3: Azure Data Factory Creation

Azure Data Factory was created to orchestrate and manage the data pipeline.

The following ADF components were explored:

- Author Hub
- Monitor Hub
- Manage Hub

Output

The first screenshot shows the 'Create Data Factory' wizard in the 'Review + create' step. The 'Basics' tab is active, displaying the following configuration:

Category	Property	Value
Basics	Subscription	Azure subscription 1
	Resource group	RG-Superstore-ADF
	Name	ADF-Superstore-Project1
	Region	Central India
Networking	Version	V2
	Connect via	Public endpoint

The 'Create' button is highlighted in blue.

The second screenshot shows the 'CreateDataFactory-20260614153438 | Overview' page. A green banner at the top states 'Deployment succeeded'. The main content area shows 'Your deployment is complete' with deployment details:

- Deployment name: CreateDataFactory-20260614153438
- Subscription: Azure subscription 1
- Resource group: RG-Superstore-ADF
- Start time: 6/14/2026, 3:39:50 PM
- Correlation ID: 464d3324-9732-4d50-a932-2cc14...

The 'Go to resource' button is visible. A sidebar on the right contains links for 'Cost management', 'Microsoft Defender for Cloud', 'Free Microsoft tutorials', and 'Work with an expert'.

The third screenshot shows the 'ADF-Superstore-Project1' resource page. The 'Overview' tab is active, displaying the following details:

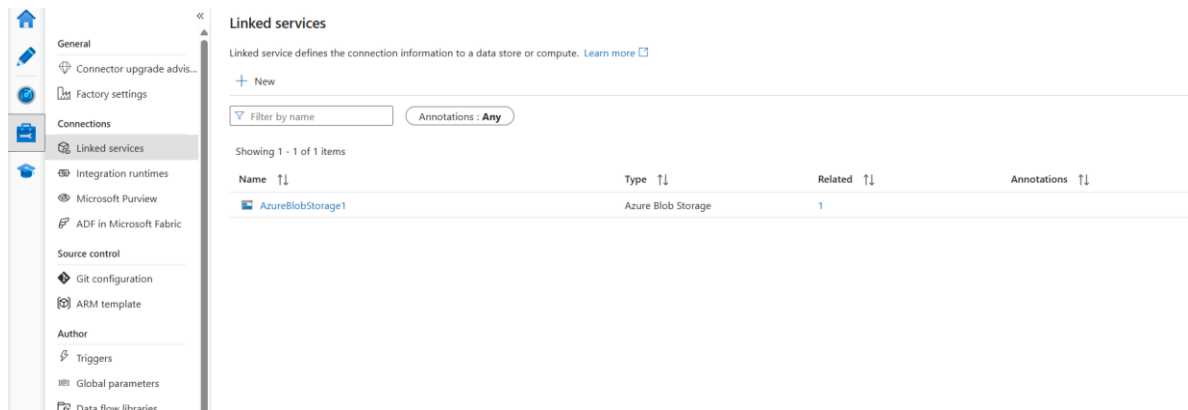
Property	Value
Resource group	RG-Superstore-ADF
Status	Succeeded
Location	Central India
Subscription	Azure subscription 1
Subscription ID	036c79c-52bf-4e95-ab90-58d8152688f5

The 'Launch studio' button is prominently displayed. Below it are links for 'Quick Starts', 'Tutorials', 'Template Gallery', and 'Training Modules'.

Step 4: Linked Service Configuration

A Linked Service was configured to establish connectivity between Azure Data Factory and Azure Blob Storage.

Output



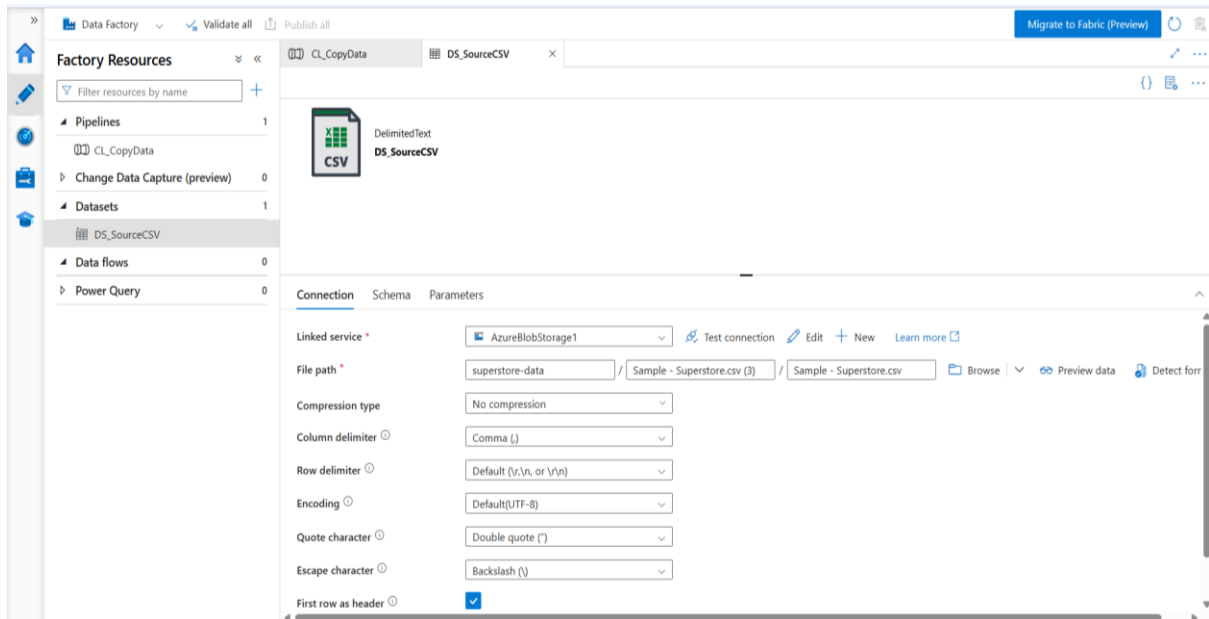
- Connection test succeeded.
- ADF successfully connected to Blob Storage.

Step 5: Dataset Creation

Source and Destination datasets were created using the uploaded CSV file.

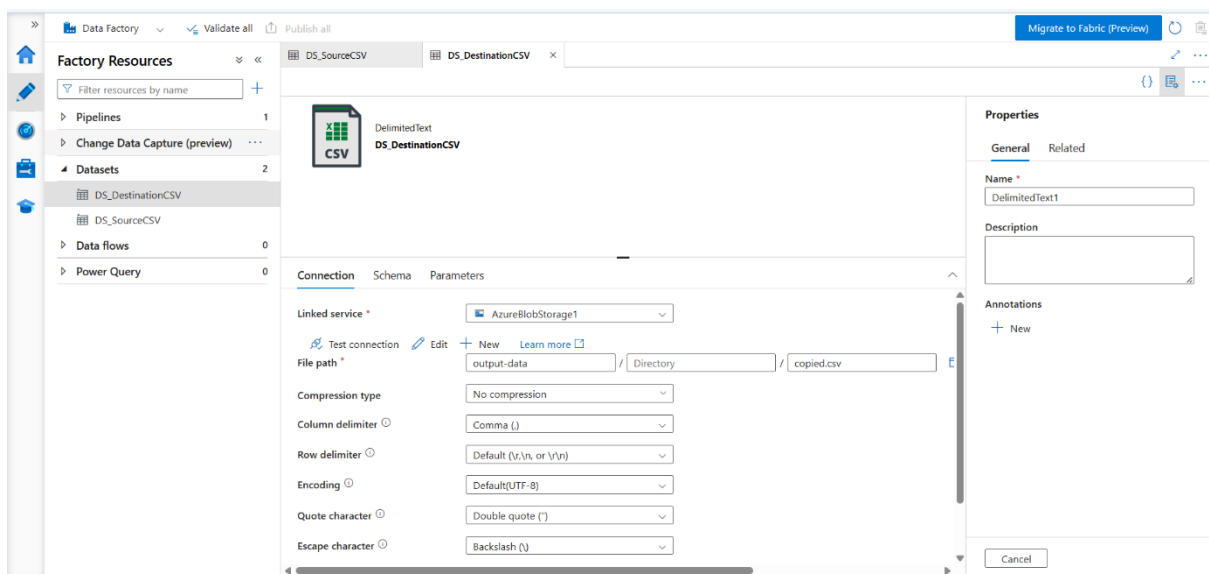
Source Dataset

- Storage Type: Azure Blob Storage
- File Name: Sample-Superstore.csv(3)



Destination Dataset

- Storage Type: Azure Blob Storage
- Target Container configured



Datasets created successfully and validated.

Step 6: Metadata Validation

The Get Metadata activity was added to validate the existence and properties of the source file before data movement.

Metadata Retrieved

Property Value

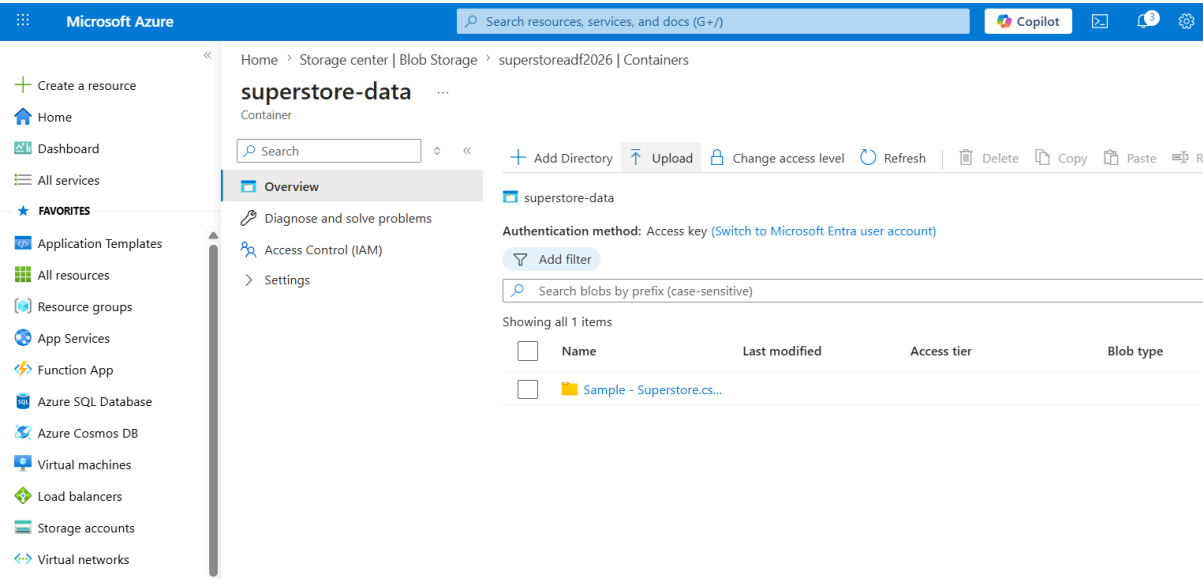
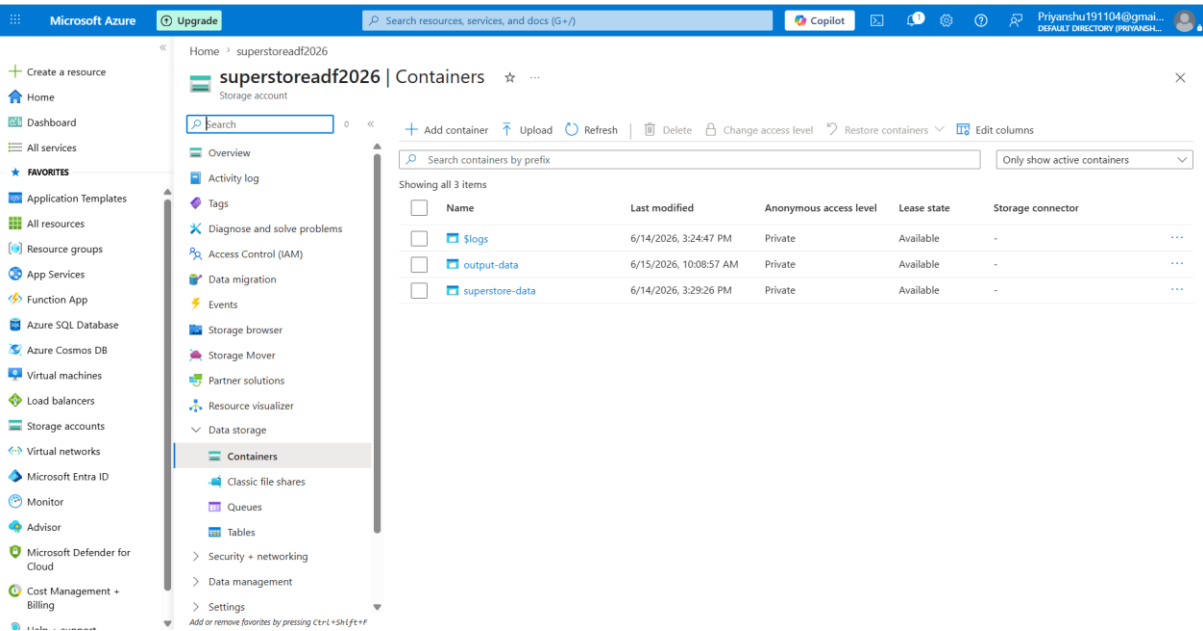
File Name Sample-Superstore.csv

File Exists True

File Type CSV

Container superstore-data

Output



Step 7: Pipeline Development

A pipeline was designed using:

1. Get Metadata Activity
2. Copy Data Activity

Pipeline Flow

SourceBlobStorage



GetMetadataActivity



CopyDataActivity



Destination Blob Storage

Output

The screenshot displays the Azure Data Factory (ADF) interface. On the left, the 'Factory Resources' pane shows a hierarchy: Pipelines (1), Change Data Capture (preview) (0), Datasets (1), Data flows (0), and Power Query (0). The 'Activities' pane lists various activity types like Lookup, Stored procedure, Script, Set variable, Validation, Web, WebHook, Wait, HDInsight, Iteration & conditionals, Machine Learning, and Power Query. The main canvas shows a pipeline named 'CL_CopyData' with two activities: 'Get Metadata' (labeled 'Get Metadata1') and 'Copy data' (labeled 'Copy data1'). Both activities are marked as successful with green checkmarks. Below the canvas, the 'Output' tab is active, showing a table of activity execution results.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	6/14/2026, 5:10:51 PM	15s	AutoResolveIntegrationRuntime (Cent
Get Metadata1	✓ Succeeded	Get Metadata	6/14/2026, 5:10:37 PM	13s	AutoResolveIntegrationRuntime (Cent

A notification box in the top right corner states: 'Running Successfully running CL_CopyData (Pipeline). View pipeline run'.

Pipeline designed successfully with proper source and destination mapping.

Step 8: Pipeline Execution

The pipeline was executed using Debug/Trigger functionality.

Execution Result

Parameter	Status
Pipeline Run	Success

Parameter	Status
Metadata Activity	Success
Copy Activity	Success
Errors	0

Output

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Dashboard

Runs

Pipeline runs

Trigger runs

Change Data Capture (previ..

Runtimes & sessions

Integration runtimes

Data flow debug

Notifications

Alerts & metrics

Dashboard

Runs

Pipeline runs

Trigger runs

Change Data Capture (previ..

Runtimes & sessions

Integration runtimes

Data flow debug

Notifications

Alerts & metrics

Pipeline runs

Triggered Debug Rerun Cancel options Refresh Edit columns List Gantt

Filter by run ID or name Chennai, Kolkata, Mu... Last 24 hours Pipeline name: All Status: All Runs: Latest runs

Triggered by: All Add filter

Showing 1 - 1 items Last refreshed 0 minutes ago

Pipeline name	Run start	Run end	Duration	Triggered by	Status	Run	Parameters
CL_CopyData	6/14/2026, 5:11:54 PM	6/14/2026, 5:12:26 PM	32s	Manual trigger	Succeeded	Original	

Run Succeeded
Successfully ran CL_CopyData (Pipeline).
[View pipeline run](#)

Step 9: Monitoring and Validation

Pipeline execution was monitored through the Monitor Hub.

Monitoring Results

Metric	Result
Pipeline Status	Succeeded
Activity Status	Succeeded
Data Transfer	Successful
Execution Time	Completed Successfully

Output

Dashboard

Runs

Pipeline runs

Monitor per runs

Change Data Capture (previ..

Runtimes & sessions

Integration runtimes

Data flow debug

Notifications

Alerts & metrics

Dashboard

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All activities completed successfully and data transfer was verified.

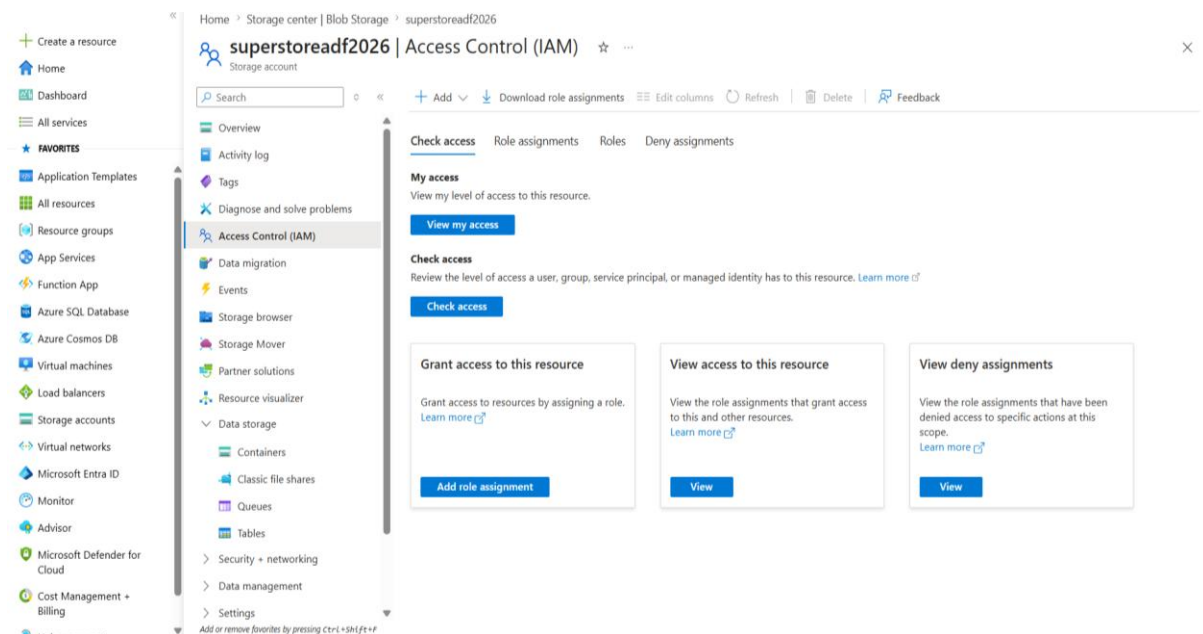
Step 10: IAM Role Assignment

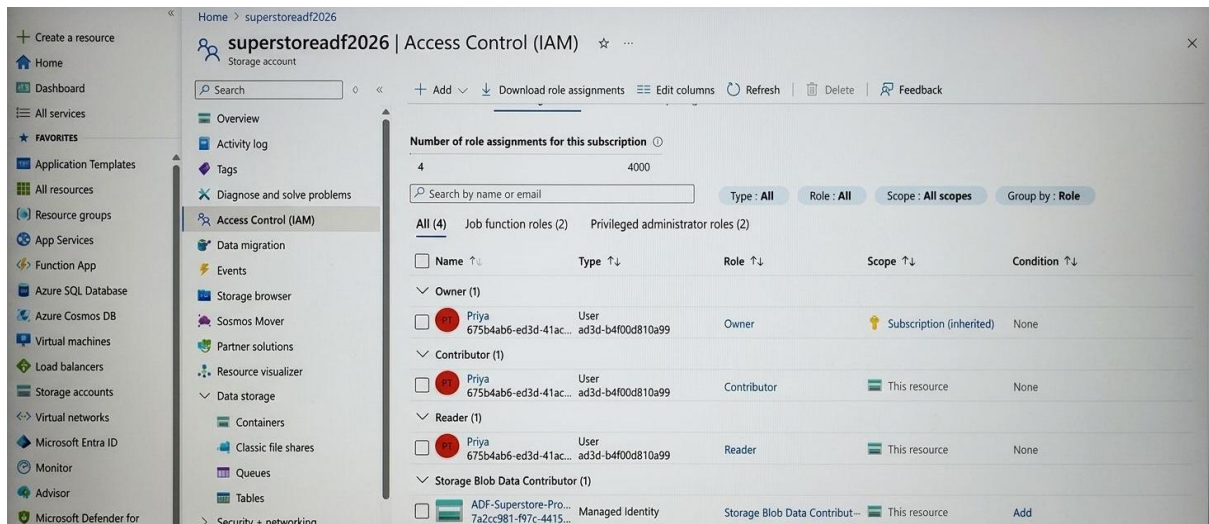
Required Azure IAM roles were assigned to ensure secure access between Azure Data Factory and Azure Storage Account.

Roles Assigned

- Reader
- Contributor
- Owner

Output





Access permissions configured successfully.

Output Summary

The Sample-Superstore.csv dataset was uploaded to Azure Blob Storage and validated using the Get Metadata activity. Azure Data Factory successfully connected to the storage account through a Linked Service and executed an end-to-end pipeline. The Copy Data activity transferred data from the source container to the destination container. Pipeline execution was monitored successfully, and all activities completed with a Succeeded status.

Conclusion

This assignment successfully demonstrated the implementation of an end-to-end Azure Data Pipeline using Azure Blob Storage and Azure Data Factory. The project covered resource provisioning, storage management, metadata validation, data movement, monitoring, and IAM configuration. The pipeline executed successfully and validated the practical use of Azure cloud services for modern data integration workflows.